Compressed Krylov and CKCI Note

April 18, 2026

Abstract

We present a compressed Krylov framework together with two practical algorithms for strongly correlated quantum many-body problems. In each Krylov step, the Hamiltonian acts on a compressed propagation vector, the resulting update is orthogonalized against the current Krylov basis, compressed under fixed propagation and storage budgets, and incorporated through a projected generalized eigenvalue problem in the resulting compressed, generally nonorthogonal Krylov space. A thick-restart variant preserves the accuracy of this budget-limited Krylov procedure by retaining several low-lying Ritz vectors between cycles, while a selected-configuration-interaction refinement built on the compressed Krylov support further reduces the resources required to reach highly accurate solutions. We demonstrate these algorithms for stretched N_2 and Cr_2 active spaces of up to 22 orbitals, and show that they recover the exact solution of the 49-qubit synthetic Hamiltonian used as a benchmark for quantum Krylov calculations in a regime where other selected configuration-interaction methods have limited reach.

1 Compressed Krylov and CKCI

We write the many-body wavefunction in a determinant basis as

$$|\Psi\rangle = \sum_I c_I |D_I\rangle, \tag{1}$$

where $|D_I\rangle$ denotes a Slater determinant (or, more generally, a configuration basis state). In the compressed Krylov (CK) algorithm, we build a sequence of sparse Krylov vectors

$$\mathcal{Q}_m = \mathrm{span}\left\{|\Psi_0\rangle, |\Psi_1\rangle, \ldots, |\Psi_{m-1}\rangle\right\}, \tag{2}$$

starting from an initial sparse seed $|\Psi_0\rangle$, typically a Hartree–Fock determinant or another guiding configuration.

At Krylov step k, the projected update is generated by Hamiltonian action on the current sparse vector,

$$|\widetilde{v}_{k+1}\rangle = H|\Psi_k\rangle. \tag{3}$$

To remove components already represented in the current Krylov basis, we form an orthogonalized residual

$$|r_{k+1}\rangle = (I - P_k)\,|\widetilde{v}_{k+1}\rangle, \tag{4}$$

where P_k denotes projection onto the span of the previously constructed Krylov vectors. In a nonorthogonal Krylov basis this projector may be written formally as

$$P_k = \sum_{i,j=0}^{k} |\Psi_i\rangle (S_k^{-1})_{ij} \langle\Psi_j|, \qquad (S_k)_{ij} = \langle\Psi_i|\Psi_j\rangle. \tag{5}$$

Because the stored Krylov vectors are compressed and generally nonorthogonal, the corresponding projected eigenvalue problem is written in the generalized Ritz form

$$H^{(m)}\mathbf{u} = E\, S^{(m)}\mathbf{u}, \tag{6}$$

with

$$H_{ij}^{(m)} = \langle\Psi_i|H|\Psi_j\rangle, \qquad S_{ij}^{(m)} = \langle\Psi_i|\Psi_j\rangle. \tag{7}$$

The associated Ritz state has the CI-like expansion

$$|\Phi^{(m)}\rangle = \sum_{k=0}^{m-1} u_k|\Psi_k\rangle. \tag{8}$$

The lowest Ritz root gives the current CK energy estimate, and the associated Ritz vector defines the best approximation within the compressed Krylov space.

Compression enters at two levels. First, only the largest-amplitude components needed for propagation are retained under a propagation budget K_{prop}. Second, the next stored Krylov vector is compressed to a storage budget K_{store}. Formally, if $\mathcal{C}_K$ denotes the operation that retains the K largest coefficients by magnitude, the stored next vector can be written schematically as

$$|\Psi_{k+1}\rangle = \frac{\mathcal{C}_{K_{\mathrm{store}}}(|r_{k+1}\rangle)}{\|\mathcal{C}_{K_{\mathrm{store}}}(|r_{k+1}\rangle)\|}. \tag{9}$$

In the thick-restart form of CK, several low-lying Ritz vectors are retained at the end of one cycle and used as restart seeds for the next cycle, rather than restarting from a single vector.

The CKCI construction uses the determinant support discovered by CK as the selected CI space. If

$$\Omega_{\mathrm{CK}} = \bigcup_{k=0}^{m-1} \mathrm{supp}(\Psi_k) \tag{10}$$

is the union of determinant supports appearing in the CK basis, then CKCI-0 is defined by exact diagonalization of the Hamiltonian projected onto this support,

$$|\Psi_{\text{CKCI-0}}\rangle = \sum_{I\in\Omega_{\mathrm{CK}}} c_I^{(0)}|D_I\rangle, \tag{11}$$

with coefficients obtained from

$$P_{\Omega_{\mathrm{CK}}} H P_{\Omega_{\mathrm{CK}}}|\Psi_{\text{CKCI-0}}\rangle = E_{\text{CKCI-0}}|\Psi_{\text{CKCI-0}}\rangle. \tag{12}$$

Higher-order CKCI refinements are obtained by enlarging the determinant support from the dominant amplitudes of the previous exactized solution. If Ω_n denotes the determinant space for CKCI-n, and $\mathcal{P}_{N_{\mathrm{top}}}(\Psi_{\text{CKCI-n}})$ denotes the set of the N_{top} determinants with the largest $|c_I^{(n)}|$, then a single shell-growth refinement may be written schematically as

$$\Omega_{n+1} = \Omega_n \cup \mathcal{N}\big(\mathcal{P}_{N_{\mathrm{top}}}(\Psi_{\text{CKCI-n}})\big)\,, \tag{13}$$

where $\mathcal{N}$ denotes the set of configurations connected by one application of H. CKCI-1 and CKCI-2 are then obtained by exact diagonalization in Ω_1 and Ω_2, respectively. In this way, CK acts as a compressed Krylov discovery stage, while CKCI provides an exactized selected-CI refinement on the determinant support uncovered by the Krylov dynamics.